

Atomic Structure and the Periodic Table

This section covers the fundamental principles of atomic structure, electron configurations, and periodic trends.

1. Which of the following describes the Aufbau principle? (A) Electrons fill orbitals of the lowest available energy levels before filling higher levels. (B) No two electrons in an atom can have the same four quantum numbers. (C) Electrons will occupy an empty orbital before they pair up in the same orbital. (D) It is impossible to know both the position and momentum of an electron simultaneously. (E) Electrons are found in specific energy levels.

- **Solution:** The **Aufbau principle** (from the German *Aufbauprinzip*, meaning "building-up principle") states that in the ground state of an atom or ion, electrons fill atomic orbitals of the lowest available energy levels before occupying higher levels. The correct option is (A).

2. Which of the following elements has the highest electronegativity? (A) F (B) Cl (C) Br (D) I (E) At

- **Solution: Electronegativity** is a measure of an atom's ability to attract shared electrons in a chemical bond. It increases across a period and decreases down a group in the periodic table. Fluorine (F), Chlorine (Cl), Bromine (Br), and Iodine (I) are all in Group 17 (the halogens). **Fluorine (F)** is at the top of the group, making it the most electronegative element. The correct option is (A).

3. Which of the following quantum numbers describes the shape of an orbital? (A) Principal quantum number (n) (B) Azimuthal quantum number (l) (C) Magnetic quantum number (m_l) (D) Spin quantum number (m_s) (E) None of the above

- **Solution:** The **azimuthal quantum number (l)**, also known as the orbital angular momentum quantum number, determines the shape of an atomic orbital.
 - $l = 0$ corresponds to an s-orbital (spherical).
 - $l = 1$ corresponds to a p-orbital (dumbbell shape).
 - $l = 2$ corresponds to a d-orbital (more complex shapes). The correct option is (B).

Chemical Bonding and Molecular Structure

These questions focus on the types of chemical bonds, molecular geometry, and hybridization theories.

4. What is the shape of a molecule with sp^3d hybridization? (A) Tetrahedral (B) Trigonal planar (C) Trigonal bipyramidal (D) Octahedral (E) Linear

- **Solution:** sp^3d hybridization involves the mixing of one s orbital, three p orbitals, and one d orbital to form five equivalent hybrid orbitals. According to VSEPR (Valence Shell Electron Pair Repulsion) theory, these five electron domains arrange themselves in a **trigonal bipyramidal** geometry to minimize repulsion. The correct option is (C).

5. Which of the following molecules has a nonpolar covalent bond? (A) H_2O (B) HCl (C) O_2 (D) NH_3 (E) CH_4

- **Solution:** A nonpolar covalent bond occurs between two atoms when the difference in their electronegativity is zero. In the **oxygen molecule (O_2)**, the bond is between two identical atoms, so there is no difference in electronegativity, and the electrons are shared equally. While CH_4 is a nonpolar molecule, the individual C-H bonds have a slight polarity. The correct option is (C).

6. Which of the following is an example of an ionic compound? (A) CO_2 (B) H_2O (C) CH_4 (D) $NaCl$ (E) O_2

- **Solution:** An **ionic compound** is formed from the electrostatic attraction between oppositely charged ions, typically formed between a metal and a non-metal. **Sodium chloride ($NaCl$)** is formed from the metal sodium (Na^+) and the non-metal chlorine (Cl^-). The other options are molecular compounds formed by covalent bonds. The correct option is (D).

Stoichiometry and Chemical Calculations

This section includes questions on mole concepts, empirical formulas, and solution concentrations.

7. A compound contains 40.0% carbon, 6.7% hydrogen, and 53.3% oxygen by mass. What is the empirical formula of the compound? (A) CHO (B) CH_2O (C) $C_2H_4O_2$ (D) $C_6H_{12}O_6$ (E) None of the above

- **Solution:**
 1. **Assume a 100 g sample:** This gives 40.0 g of Carbon, 6.7 g of Hydrogen, and 53.3 g of Oxygen.
 2. **Convert mass to moles:**
 - C: $40.0\text{ g}/12.01\text{ g/mol}=3.33\text{ mol}$

- H: $6.7 \text{ g} / 1.01 \text{ g/mol} = 6.63 \text{ mol}$
 - O: $53.3 \text{ g} / 16.00 \text{ g/mol} = 3.33 \text{ mol}$
3. **Find the simplest whole-number ratio:** Divide all mole values by the smallest value (3.33).
- C: $3.33 / 3.33 = 1$
 - H: $6.63 / 3.33 \approx 2$
 - O: $3.33 / 3.33 = 1$ The empirical formula is **CH₂O**. The correct option is (B).

8. What is the molarity of a solution prepared by dissolving 40.0 g of NaOH in enough water to make 500.0 mL of solution? (A) 0.5 M (B) 1.0 M (C) 2.0 M (D) 4.0 M (E) 8.0 M

• **Solution:**

1. **Calculate the molar mass of NaOH:** $23.0 + 16.0 + 1.0 = 40.0 \text{ g/mol}$.
2. **Calculate the moles of NaOH:** Moles = mass / molar mass = $40.0 \text{ g} / 40.0 \text{ g/mol} = 1.0 \text{ mol}$.
3. **Convert volume to liters:** $500.0 \text{ mL} = 0.500 \text{ L}$.
4. **Calculate molarity:** Molarity = moles / volume (L) = $1.0 \text{ mol} / 0.500 \text{ L} = 2.0 \text{ M}$. The correct option is (C).

Gases and Chemical Reactions

This part covers the behavior of gases and the classification of chemical reactions.

9. A gas occupies 2.0 L at 1.0 atm. What volume will it occupy at 4.0 atm, assuming constant temperature? (A) 0.5 L (B) 1.0 L (C) 2.0 L (D) 4.0 L (E) 8.0 L

- **Solution:** This problem uses **Boyle's Law**, which states that for a fixed amount of gas at constant temperature, pressure and volume are inversely proportional ($P_1V_1 = P_2V_2$).
Given: $P_1 = 1.0 \text{ atm}$, $V_1 = 2.0 \text{ L}$, $P_2 = 4.0 \text{ atm}$. $(1.0 \text{ atm})(2.0 \text{ L}) = (4.0 \text{ atm})(V_2)$
 $V_2 = 4.0 \text{ atm} \cdot 2.0 \text{ L} / 4.0 \text{ atm} = 0.5 \text{ L}$. The correct option is (A).

10. Which of the following is a decomposition reaction? (A) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ (B) $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$ (C) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (D) $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$ (E) None of the above

- **Solution:** A **decomposition reaction** is a type of reaction in which one compound breaks down into two or more simpler substances. The reaction $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ shows a single compound, calcium carbonate, breaking down into calcium oxide and carbon dioxide. The correct option is (C).